

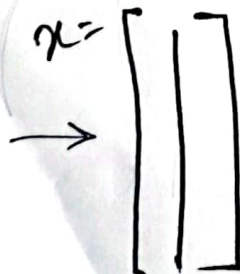
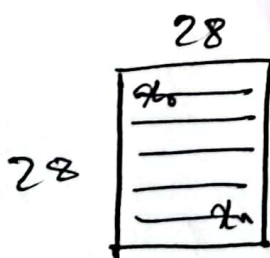
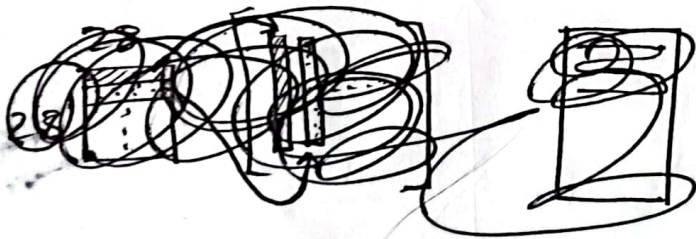
Classification Problem

y is discrete

e.g. Binary $\rightarrow y = \{0, 1\}$

x_1	x_2	x_3	y

Example: image pixels $\rightarrow$ cat/dog

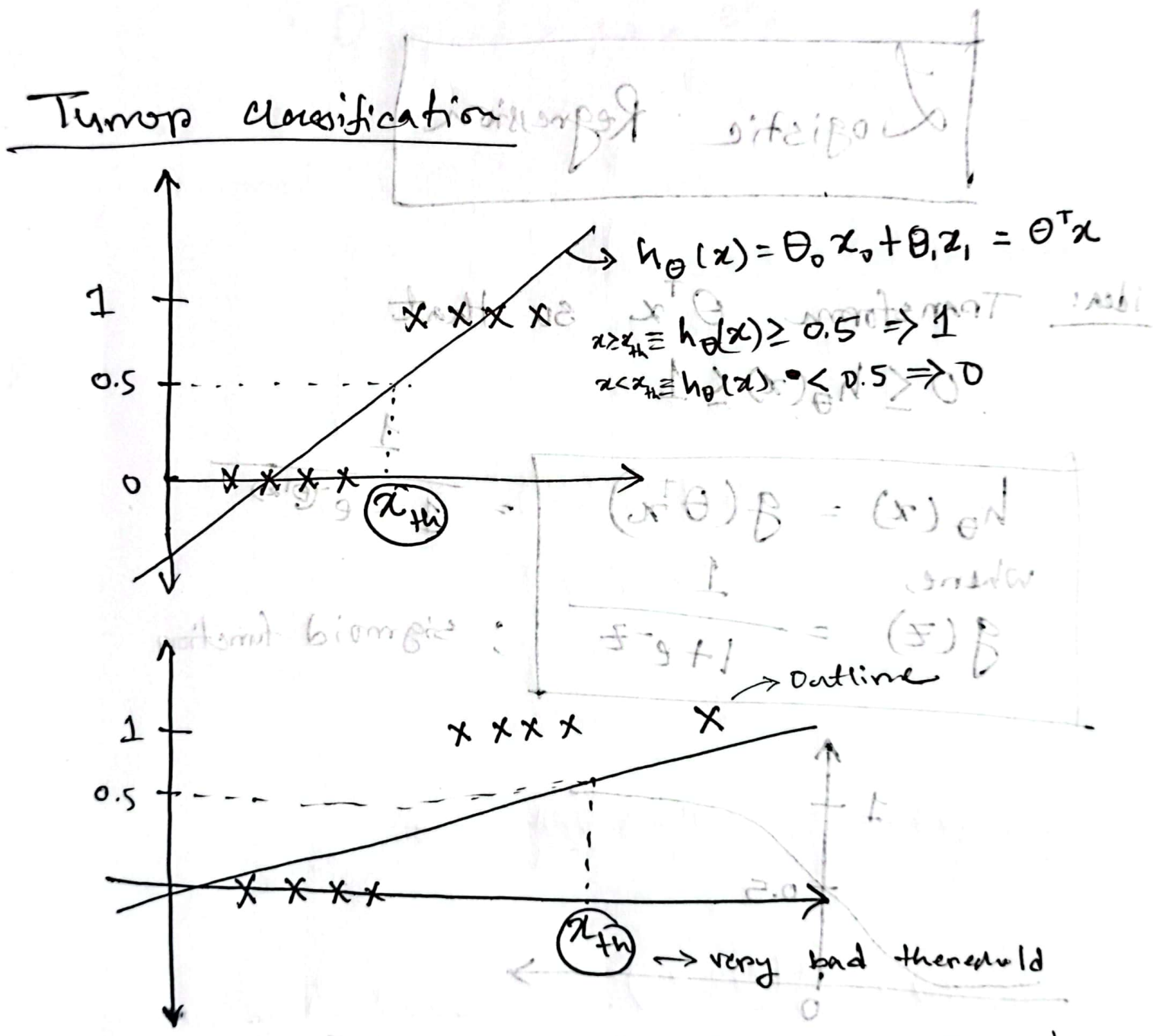


একটি image এর
২৪ pixel একটি
column এ থাকবে।

$$x = [x_0, x_1, \dots, x_n]$$

Example: tumor size $\rightarrow 0/1$

$x = [x_0, x_1]$



Classification problem - (2): Linear regression

→ ① sensitive to outline

② $y = 0$ or 1

$h_\theta(x) = \theta^T x$ can be $< 0, > 1$

x_{th} দিয়ে ০/১ বানানো লাগে.

So ~~we~~ we need a dedicated classification algo.

Logistic Regression

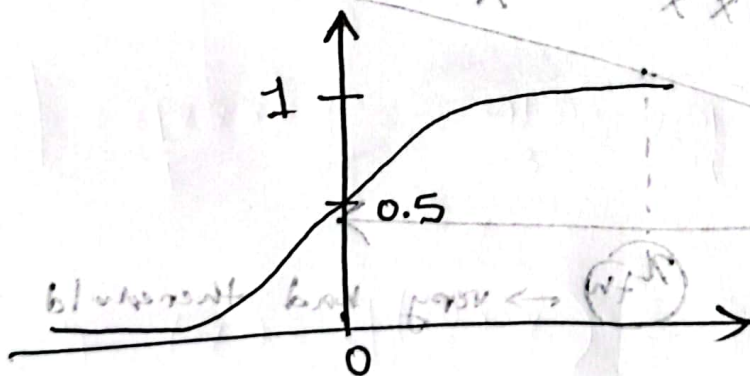
idea: Transform $\theta^T x$ so that
 $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = g(\theta^T x)$$

where,

$$g(z) = \frac{1}{1 + e^{-z}}$$

; Sigmoid function



$$\hat{y} = \begin{cases} 1 & \text{if } h_{\theta}(x) \geq 0.5 \\ 0 & \text{if } h_{\theta}(x) < 0.5 \end{cases}$$

$$y = \begin{cases} 1 & \text{if } \frac{1}{1 + e^{-(\theta^T x)}} \geq 0.5 \\ 0 & \text{if } \frac{1}{1 + e^{-\theta^T x}} < 0.5 \end{cases}$$

interpret: $h_{\theta}(x) = P(y=1 | x, \theta)$

Probability of y being 1



$20 \leq (x) \leq 1$
 $0 \leq y \leq 1$
 $20 > (x) \leq 1$
 $0 \geq y \leq 1$



$$(x, \theta + \delta \theta) \cdot \beta = (x^T \theta) \beta = (x) \text{ on}$$

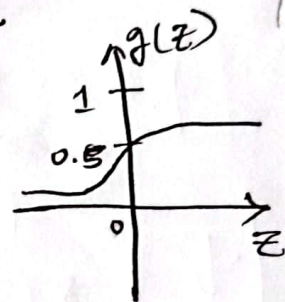
$$\begin{cases} x + \delta x = x^T \theta \\ x + \delta x = \begin{bmatrix} 1 \\ x \end{bmatrix} = \begin{bmatrix} \theta \\ \theta \end{bmatrix} = \theta \end{cases}$$

Decision Boundary

$h_{\theta}(x) = g(\theta^T x) = g(z)$

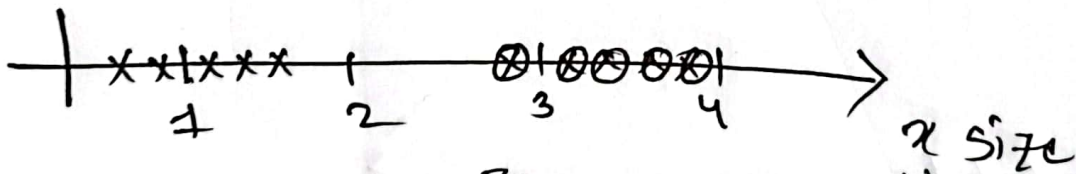
$(\theta, x | 1 = 0) \text{ if } z = 0$

$\frac{1}{1 + e^{-\theta^T x}}$



$$y = \begin{cases} 1 & \text{if } h_{\theta}(x) \geq 0.5 \\ & \hookrightarrow z \geq 0 \\ 0 & \text{if } h_{\theta}(x) < 0.5 \\ & \hookrightarrow z \leq 0 \end{cases}$$

Ex 1



$h_{\theta}(x) = g(\theta^T x) = g(\theta_0 + \theta_1 x)$

$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ x \end{bmatrix}$

$\theta^T x = \theta_0 + \theta_1 x$
 $= -2 + x$

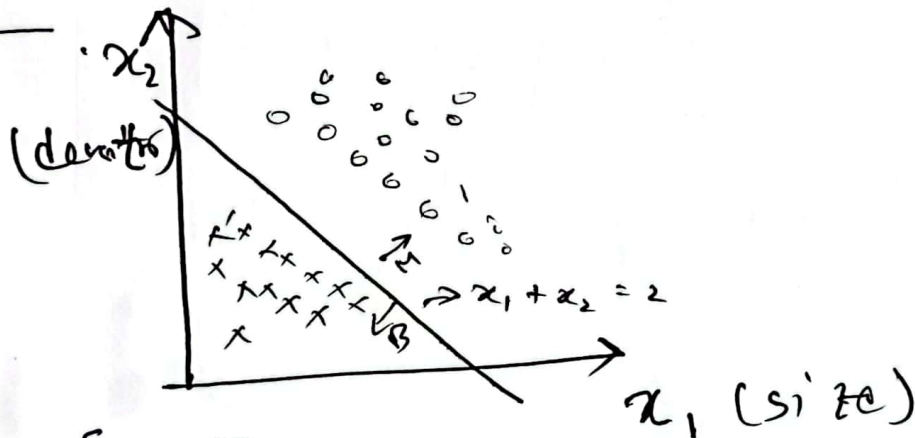
Decision boundary:

$$z \geq 0$$

$$\Rightarrow -2 + x \geq 0$$

$$\boxed{x \geq 2}$$

Ex 2



$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} \xrightarrow{\text{after op}} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore z = \theta^T x = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \\ = -2 + x_1 + x_2$$

$$\text{Now, } z \geq 0$$

$$\Rightarrow -2 + x_1 + x_2 \geq 0$$

$$\therefore x_1 + x_2 \geq 2$$

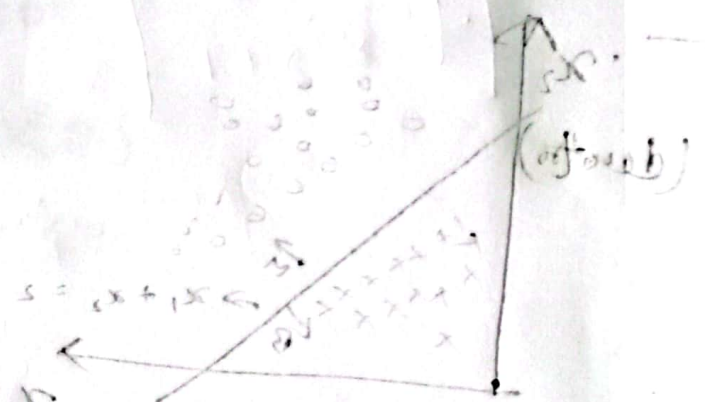
Ex: 3

Decision boundary

$$0 \leq f$$

$$0 \leq x + s - \leq$$

$$[s \leq x]$$



$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \cdot \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = f$$

$$x_1 \theta_1 + x_2 \theta_2 + \theta_0 = x' \theta = f$$

$$-s + x_1 + x_2$$

$$0 \leq f$$

$$0 \leq x + s - \leq$$

$$s \leq x + x_1$$

Fitting the Model $\equiv$ Finding the best θ (Learning Algo)

$$D = \left\{ (x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)}) \right\}$$

$$X = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{bmatrix}$$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

[single x]

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_d \end{bmatrix}$$

$$z = \theta^T x$$

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

Task: find the "best" θ

θ that performs best on dataset, D

$\hookrightarrow$ minimizes cost function, $J(\theta)$

(cost function) θ fed into gradient = global min gradient

Cost Function: $J(\theta) = \frac{1}{n} \sum_{i=1}^n J_i(\theta)$

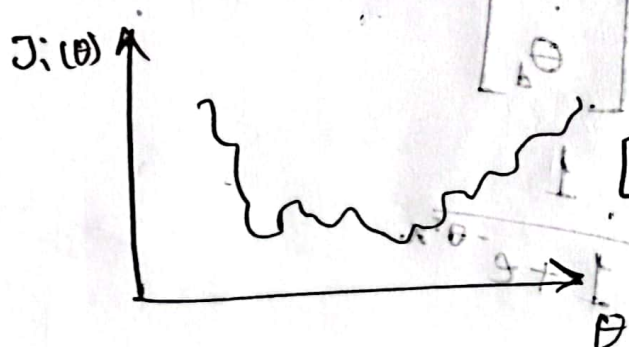
$\left\{ \begin{matrix} \text{Avg} \\ \text{Cost} \end{matrix} \right\}$ $\left\{ \begin{matrix} \text{Cost for single} \\ \text{ith sample} \end{matrix} \right\}$

Option 1: MSE: $J_i(\theta) = [h_{\theta}(x) - y]^2$

$= \left[\frac{1}{1 + e^{-\theta^T x}} - y \right]^2$

$x^T \theta = \theta_0 + \theta_1 x_1 + \dots + \theta_n x_n$

Problem: not convex (V)



[many local minima]

(1) for global min gradient
 (2) for local min gradient

option 2

Binary Cross entropy loss

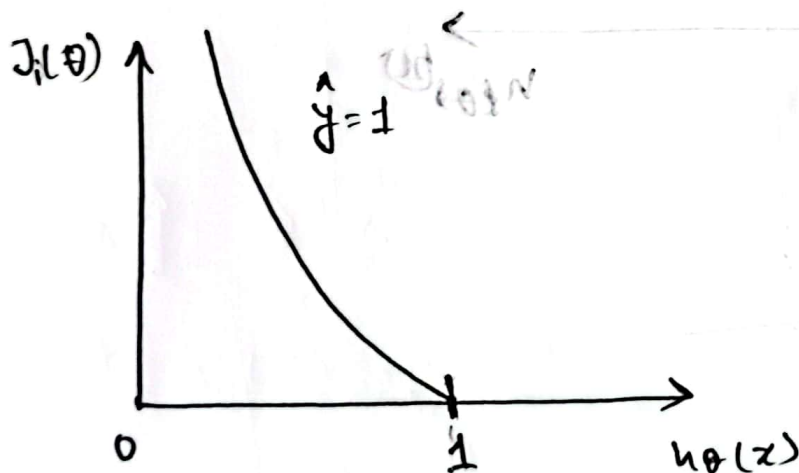
log base = e

$$J_i(\theta) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

output of model

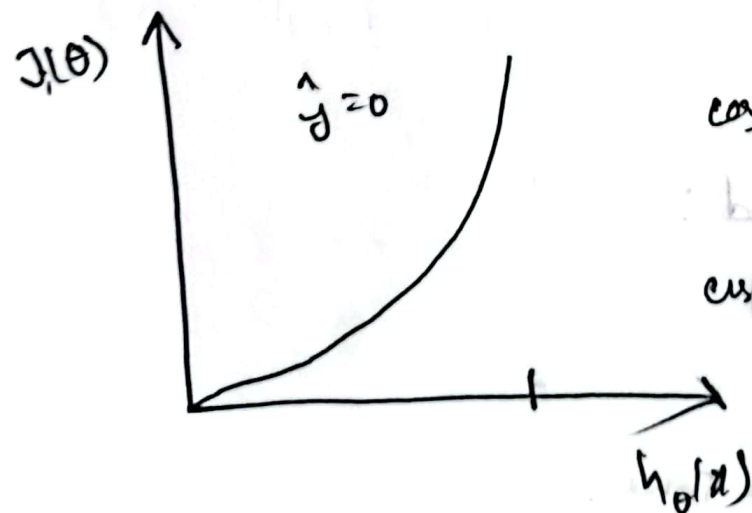
cost function for its sample

$$J_i(\theta) = -[y \times \log(h_\theta(x)) + (1-y) \log(1 - h_\theta(x))]$$



cost, $J(\theta) = 0$ if $\hat{y} = 1$ and $h_\theta(x) = 1$

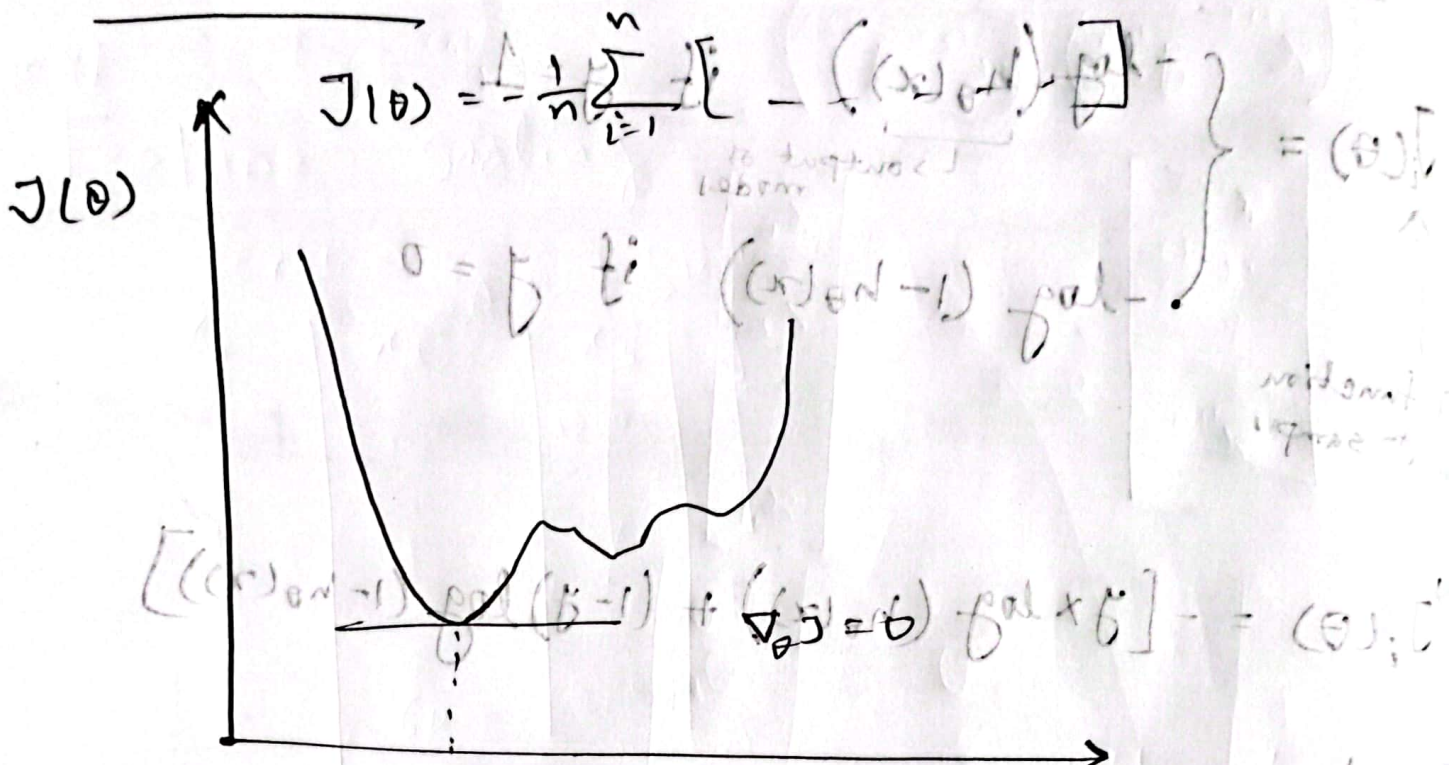
cost $J(\theta) = \infty$ if $\hat{y} = 1$ and $h_\theta(x) = 0$



cost $J(\theta) = 0$, if $\hat{y} = 0$ and $h_\theta(x) = 0$

cost $J(\theta) = \infty$, if $\hat{y} = 0$ and $h_\theta(x) = 1$

an AvB:



Minimizing the Cost:

Minimizing the Cost:

Optimizer

① Gradient descent

$\theta^0 = \text{random}$

while not converged:

$$\theta^{k+1} = \theta^k - \alpha \nabla_{\theta} J$$

if $\nabla_{\theta} J \approx 0$:

Converged = True.

- ④ LGBTQ

prilac gortat

1500 - 2000

(m) 2012

1500	2000	2012
1	1	1
2	2	2
3	3	3
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98	98	98
99	99	99
100	100	100

konstlib class ATmp; constref type

$$1 \geq x \geq 1 - \epsilon$$

Practical Considerations

**

Feature Scaling

x_1 $\rightarrow$ size (cm)	x_2 $\rightarrow$ Age	x_3 $\rightarrow$ 1200 - 5000	y
3	10	1200	
4	...	...	
5	...	...	
8	60	5000	

ଏହା Feature ମାନର scale different.

[Feature scaling କରିବା হলো সব ফিচারকে
একই scale এ নিয়ে আনা without
changing the relation between the
features.

[best option • হলো $-1 \leq x_i < 1$

[for image $(0-255) \rightarrow (0-1)$

2 or popular option: for normalization:

① Min max normalization:

$$x_n^{(i)} = \frac{x_n - \min(x_n)}{\max(x_n) - \min(x_n)} \Rightarrow 0 < x_n < 1$$

② Standardization:

$$x_n^{(i)} = \frac{x_n - \text{mean}(x^{(i)})}{\text{std}(x^{(i)})}$$



$$\frac{x_n^{(i)} - \text{mean}(x^{(i)})}{\max(x^{(i)}) - \min(x^{(i)})}$$

** Hyper-parameter selection:

Parameter, $\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$

→ which $J(\theta)$?

→ which features? $\begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$ or $\begin{bmatrix} x_0 \\ x_1 \\ x_1^2 \\ x_2 \end{bmatrix}$ or $\begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$

→ which optimizer?

→ ~~How~~ How many epoch?

Lecture - 11

P1

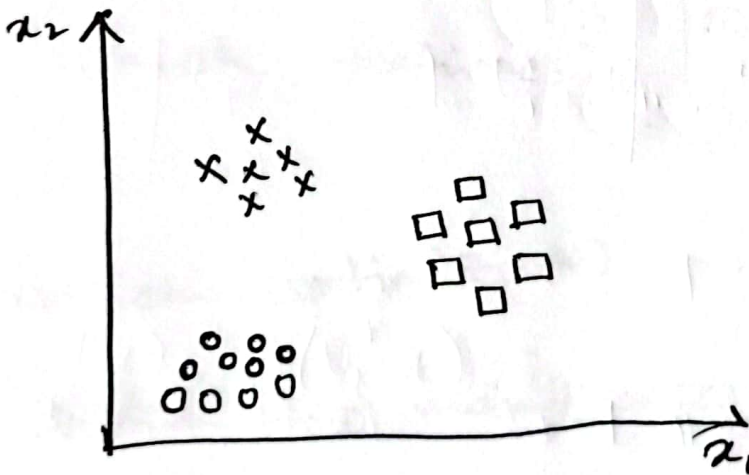
Rev.

P2

** Multiclass classification

Email : ⁰ Friends / ¹ Colisue / ² Spam

Image : ⁰ human / ¹ horse / ² cat / ³ dog



$$y = \{0, 1, 2\}$$

o x □

** Algo - 1

One vs all

$$y = \{0, 1, 2\}$$

Idea: Train a logistic regression classifier

(i) h_{θ_i} for each class i to predict the probability of $y = i$

3 θ create

$$(h_{\theta_0}) = \theta_0^T x$$

$$\{\theta_{00}, \theta_{01}, \dots, \theta_{0d}\}$$

এক input ফিল
output ফিল

~~এক data x~~

এক data x

'0' এর
probability

বাক

$$(h_{\theta_1}) = \theta_1^T x$$

$$\{\theta_{10}, \theta_{11}, \dots, \theta_{1d}\}$$

এক input ফিল

এক data x

'1' এর
probability

বাক

$$(h_{\theta_2}) = \theta_2^T x$$

$$\{\theta_{20}, \theta_{21}, \dots, \theta_{2d}\}$$

Data x এর

'2' এর

probability

বাক

বাক

$$x^T \theta + 1 = (x^T \theta) + 1 = (x^T \theta) + 1$$

$$\text{So } h_{\theta_0}^{(0)} = 0.2$$

$$h_{\theta_1}^{(1)} = 0.1$$

$$h_{\theta_2}^{(2)} = 0.9$$

যদি probability তাকি

① Data x ক class 0

বাক

$$y_{pred} = \arg \max_i h_{\theta_i}^{(i)}(x)$$

$$1 - \log 4$$

$$h_{\theta_0}^{(0)}(x) = g(\theta_0^T x) = \frac{1}{1 + e^{-\theta_0^T x}} = P(y=0 | x, \theta_0)$$

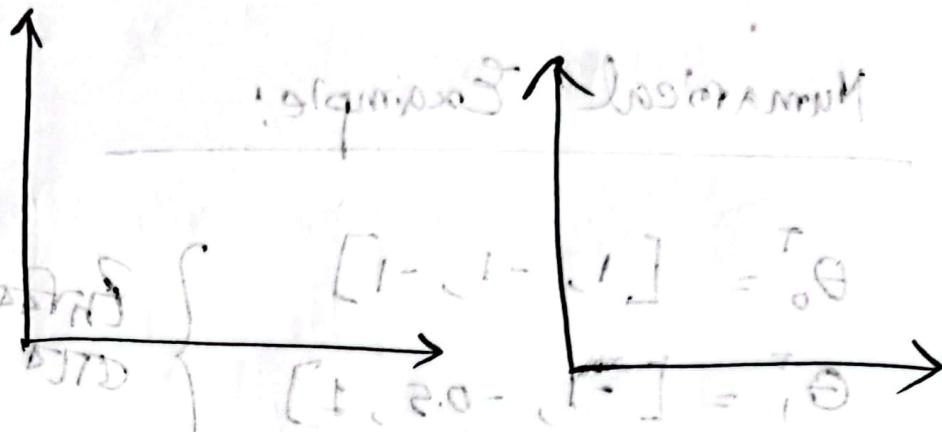
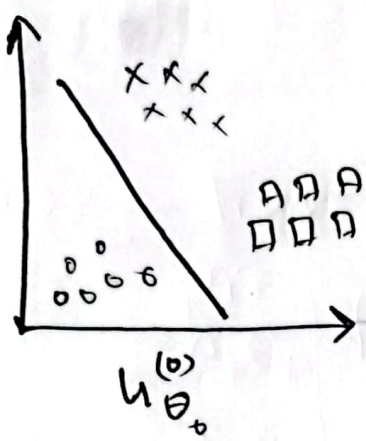
$$[\theta_{00}, \theta_{01}, \theta_{02}, \dots, \theta_{0d}]$$

$$h_{\theta_1}^{(1)}(x) = g(\theta_1^T x) = \frac{1}{1 + e^{-\theta_1^T x}} = P(y=1 | x, \theta_1)$$

$$[\theta_{10}, \theta_{11}, \theta_{12}, \dots, \theta_{1d}]$$

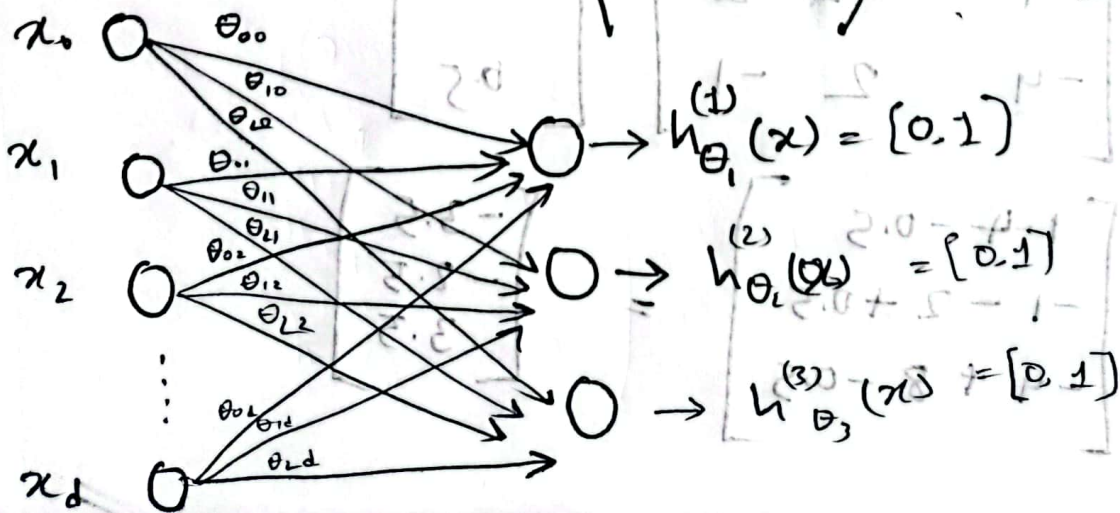
$$h_{\theta_2}^{(2)}(x) = g(\theta_2^T x) = \frac{1}{1 + e^{-\theta_2^T x}} = P(y=2 | x, \theta_2)$$

$$[\theta_{20}, \theta_{21}, \theta_{22}, \dots, \theta_{2d}]$$



$$W = \begin{bmatrix} \theta_0^T \\ \theta_1^T \\ \theta_2^T \end{bmatrix} = \begin{bmatrix} \theta_{00}, \theta_{01}, \dots, \theta_{0d} \\ \theta_{10}, \theta_{11}, \dots, \theta_{1d} \\ \theta_{20}, \theta_{21}, \dots, \theta_{2d} \end{bmatrix}$$

$$h_{\theta}(x) = g(Wx) = g \left(\begin{bmatrix} \theta_0^T x \\ \theta_1^T x \\ \theta_2^T x \end{bmatrix} \right) = \begin{bmatrix} g(\theta_0^T x) \\ g(\theta_1^T x) \\ g(\theta_2^T x) \end{bmatrix}$$



Numerical Example:

$$\theta_0^T = [1, -1, -1]$$

$$\theta_1^T = [-1, -0.5, 1]$$

$$\theta_2^T = [-4, 2, -1]$$

ইতি
এক প্রশ্ন

$$x = \begin{bmatrix} 1 \\ 4 \\ 0.5 \end{bmatrix}$$

$$W = \begin{bmatrix} 1 & -1 & -1 \\ -1 & -0.5 & 1 \\ -4 & 2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \theta_0^T \\ \theta_1^T \\ \theta_2^T \end{bmatrix}$$

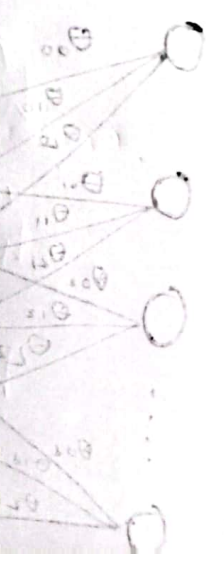
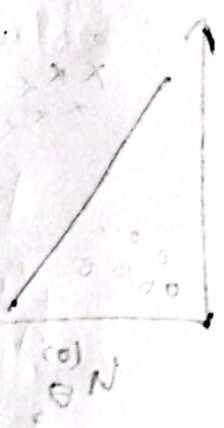
$$Z = Wx$$

$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & -0.5 & 1 \\ -4 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 0.5 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 4 \\ 0.5 \end{bmatrix}$$

$$\begin{bmatrix} 1 - 4 - 0.5 \\ -1 - 2 + 0.5 \\ -4 + 8 - 0.5 \end{bmatrix}$$

$$\begin{bmatrix} -3.5 \\ -2.5 \\ 3.5 \end{bmatrix}$$



$$g(z) =$$

$$\begin{bmatrix} \frac{1}{1+e^{3.5}} \\ 1 \\ \frac{1}{1+e^{2.5}} \\ 1 \\ \frac{1}{1+e^{3.5}} \end{bmatrix}$$

$$= \begin{bmatrix} 0.03 \\ 0.08 \\ 0.97 \end{bmatrix}$$

$$\therefore \hat{y}_{pred} = 2$$

How

to get

$$\theta_0, \theta_1, \theta_2?$$

$$\text{Parameters} = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}^{(0)}, \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}^{(1)}, \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}^{(2)}$$

Then optimize by : G.D / Adam / ...

** Problems - of one vs all classifier:

① Have to train separately

② No probabilistic interpretation for h !
[sum(h) > 1]

(যদি $\text{sum} = 1$ হয় তা,
যদি each class এর probability হয়
তাহলে $\text{sum} = 1$ হবে)

** Solution: { ① we ~~must~~ want $\text{sum}(h) = 1$
② we want end to end learning

Softmax Classifier

⇒ modification to the Previous algorithm:

~~① Sigmoid function to softmax function~~

① Sigmoid function to softmax function

② change z to one-hot vector.

③ from $J(\theta) = \text{BCE}$ to CE
Binary Cross Entropy Cross Entropy

① $\Rightarrow$

~~z~~

$$Wx = [z_0, z_1, \dots, z_{k-1}]$$

$\downarrow$ $\downarrow$
 $(k \times d+1)$ $(d+1 \times 1)$
 $\downarrow$ $\downarrow$
 $(k \times 1)$

$$\text{sigmoid}(Wx)^T = \left[\frac{1}{1+e^{-z_0}}, \frac{1}{1+e^{-z_1}}, \dots \right]$$

Softmax function, $f_i = \frac{e^{z_i}}{\sum_{i=0}^{k-1} e^{z_i}}$

~~sigmoid~~

$$\text{Softmax}(Wx)^T = [f_0, f_1, \dots, f_{k-1}]$$

$$f_1 = \frac{e^{z_1}}{e^{z_0} + e^{z_1} + e^{z_2}}$$

$$\sum_{i=0}^{k-1} f_i = 1$$

② $\Rightarrow$ one hot ~~vector~~ encoding

$$y = \{0, 1, 2\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \text{①}$$

③ $\Rightarrow$ $\begin{bmatrix} \frac{1}{1+e^{-x_0 w}} & \frac{1}{1+e^{-x_1 w}} \end{bmatrix} = \sigma(xw)$ logistic

cross entropy loss

$$J(w) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=0}^{K-1} y_k^{(i)} \log f_k$$

$\downarrow$
 $J_i(w)$

Optimizer $\Rightarrow$

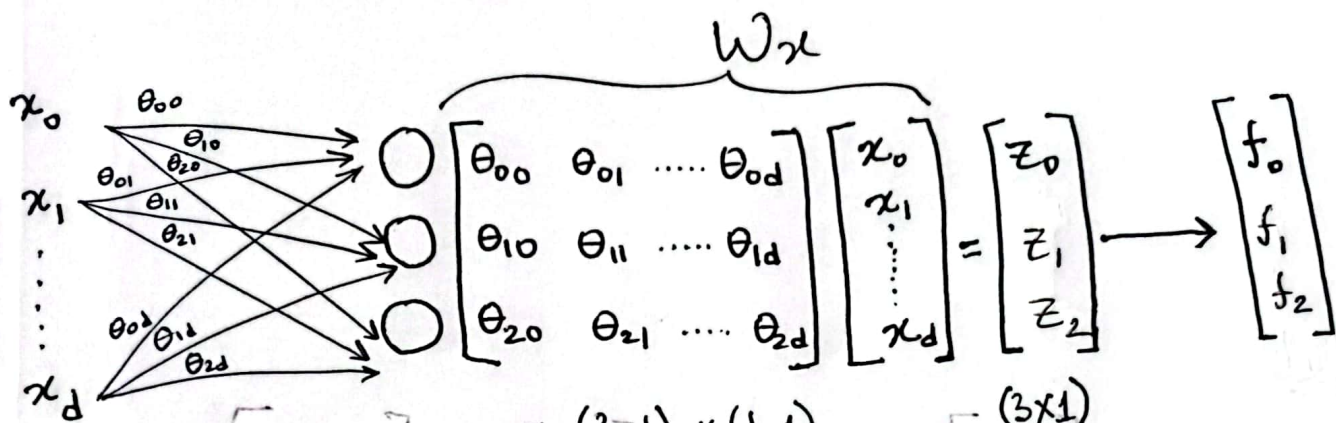
$w^0 \Leftarrow$ random

while not converged:

$$w^{(k+1)} = w^k - \nabla_w J$$

if $\nabla_w J \approx 0$:

converged = True



$$\theta_0 = \begin{bmatrix} 0.10 & 1.82 & 0.35 & 0 \\ 0 & 0.01 & -0.05 & 0.1 & 0.05 \end{bmatrix}$$

$$\theta_1 = [0.2, 0.7, 0.2, 0.05, 0.16]$$

$$\theta_2 = [-0.3, 0.0, -0.45, -0.2, 0.33]$$

$$x_1 = [1.0, 0.15, 1.22, 0.44, 0.56], d = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$z_0 = 0 \times 1 - 0.01 \times 15 - 0.05 \times 22 - 0.1 \times 44 + 0.05 \times 56 = -2.58$$

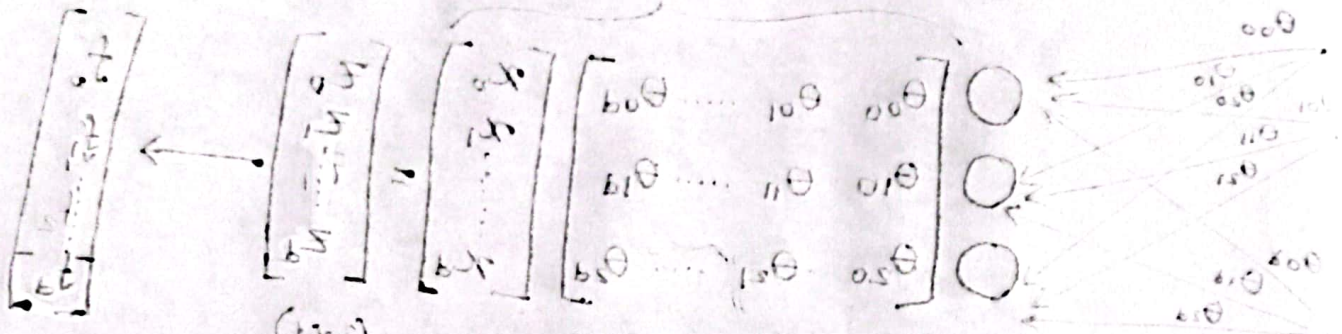
$$z_1 = 0.02 \times 1 - 0.7 \times 15 + 0.2 \times 22 - 0.05 \times 44 + 0.16 \times 56 = 0.86$$

$$z_2 = -0.3 \times 1 - 0 \times 15 - 0.45 \times 22 + 0.2 \times 44 + 0.33 \times 56 = 0.28$$

$$\therefore \begin{bmatrix} z_0 \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} -2.58 \\ 0.86 \\ 0.28 \end{bmatrix}$$

$$\therefore f \begin{bmatrix} z_0 \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \end{bmatrix}$$

$$f = \begin{bmatrix} 0.016 \\ 0.631 \\ 0.353 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$



Now,

$$y = \begin{bmatrix} 0 \\ 0 \\ 1.0 \end{bmatrix}$$

$$y_{\text{pred}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0.016 \\ 0.631 \\ 0.323 \end{bmatrix}$$

$$J_d^{(w)} = - \sum_{k=0}^{K-1} y_k \log f_k$$

$$= -(0 \times \log(0.016) + 0 \times \log(0.631) + 1 \times \log(0.323))$$

$$= -\log(0.323) \approx 1.04$$

(Suppose)

$$\therefore x_1 \text{ is correct, } J_1(w) = 2.06$$

$$\therefore x_2 \text{ is correct, } J_2(w) = 0.08$$

$$\therefore x_3 \text{ is correct, } J_3(w) = 1.34$$

$$J(w) = \frac{1}{n} \sum_{i=0}^{n-1} \left(- \sum_{k=0}^{K-1} y_k \log f_k \right)$$

$$= \frac{1}{4} [J_0(w) + J_1(w) + J_2(w) + J_3(w)]$$

$$= \frac{1}{4} (1.04 + 2.06 + 0.08 + 1.34) = 1.13$$

$n \rightarrow$ number of samples
 $K \rightarrow$ number of class